Advanced Classical Mechanics

Lecture 9: Lagrangian Dynamics

Readings: Morin - Chapter 6, Marion Chapter 7

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Hamilton's Principle

The Newtonian equation $\boldsymbol{F} = \dot{\boldsymbol{p}}$ can be readily applied to simple problems in Cartesian coordinates to describe the dynamics of classical physical systems. But as soon as we deal with non-cartesian coordinates or have ***forces of constraints*** (e.g., forces that bound objects to move on non-planar surfaces), the problem becomes too complex to describe using the Newtonian ***differential forms*** of equations.

Hamilton's principle unifies many independent observations and similar principles that aim to describe physical systems using an integral approach (rather than the differential equations approach of Newtonian dynamics). It is more general than Newtonian dynamics and applicable to a wide range of problems in Physics beyond classical mechanics, including Quantum Mechanics and Quantum Field Theory.

Hamilton's principle is based on the empirical observation that nature always minimizes certain important quantities when a physical process takes place:

> *Of all the possible paths along which a dynamical system may move from one point to another within a specified time interval (consistent with any constraints),* ***the actual path followed minimizes the time integral of the difference between the kinetic and potential energies:***

$$\delta \int_{t_1}^{t_2} (T - U)\, dt = 0 = \int_{t_1}^{t_2} L\big(x_i(t), \dot{x_i}(t), t\big)\, dt$$

where the quantity L is called the **Lagrange function** or the ***Lagrangian***. The Euler-Lagrange equations corresponding to Hamilton's principle are known as **Lagrange's equations of motion** for a particle,

$$\frac{\partial f}{\partial x_i} - \frac{d}{dt}\frac{\partial f}{\partial \dot{x}_i} = 0$$

Within the realm of Classical Mechanics, **Hamilton's principle** offers no new physics beyond Newton's laws. One can take Newton's law of motion as a principle and derive Hamilton's principle, or vice versa.

Lagrange's equations of motion

Problem 1. (Newton-Lagrange's equation equivalence) We have emphasized from the outset that the Lagrangian and Newtonian formulations of mechanics are equivalent: The viewpoint is different, but the content is the same. Starting from Lagrange's equations show that the two sets of equations of motion are, in fact, the same, by deriving the Newton's equation $\boldsymbol{F} = \dot{\boldsymbol{p}}$.

Lagrangian vs. Newtonian Dynamics

- Whereas the Newtonian approach emphasizes an outside agency acting on a body (the force), the Lagrangian method deals only with quantities associated with the body (the kinetic and potential energies). **Nowhere in the Lagrangian formulation does the concept of force enter.**

- Because the Lagrangian depends only on the scalar quantities (energy), it is invariant under a coordinate transformation. This allows the use of Lagrangian outside of Classical Mechanics (e.g., Quantum Mechanics) , where systems are typically described by their energies (as opposed to external forces acting on them).

Problem 2. (Generalized Coordinates) Suppose we have a set of coordinates $q_i = q_i(x_1, x_2, \ldots, x_N;\ t)$ where t is time. Show that the original Lagrange's equations for the Cartesian coordinates seen in the previous slides can be also written as,

$$\frac{\partial f}{\partial q_i} - \frac{d}{dt}\frac{\partial f}{\partial \dot{q}_i} = 0$$

Problem 3. (Linear momentum conservation) Write down the Lagrangian for a frictionless ball thrown in the air in an arbitrary direction on earth. Show that the lack of dependence of the Lagrangian on the x and y coordinates implies that the momenta along the x and y directions are conserved.

Problem 4. (Angular and linear momentum in cylindrical coordinates) Consider a potential that depends only on the distance to the z axis. Show that the lack of dependence of the Lagrangian on the z axis implies that the linear and angular momenta along the z axis are conserved.

Lagrange's equations of motion: Noether's Theorem

Problem 5. (Angular momentum in spherical coordinates) In spherical coordinates, consider a potential that depends only on r and θ, where θ is the angle down from the north pole. Show that Lagrange's equations lead to the conservation of angular momentum around the z-axis.

Problem 6. (Conservation of Energy) In the above problems, we saw that the conservation of momentum or angular momentum arose when the Lagrangian was independent of x, y, z, θ, or φ. Show that the law of **conservation of energy arises when the Lagrangian is independent of time**. **In other words, the Lagrangian of a close (isolated) system does not *explicitly* depend on time.**

Noether's Theorem is among the most beautiful and useful theorems in physics. It deals with two fundamental concepts, namely symmetries and conserved quantities.

For each symmetry of the Lagrangian, there is a conserved quantity.

By "symmetry," we mean that if the coordinates are changed by some small quantities, then the Lagrangian has no first-order change in these quantities. By "conserved quantity," we mean a quantity that does not change with time.

We have already seen this theorem in effect in the previous problems.

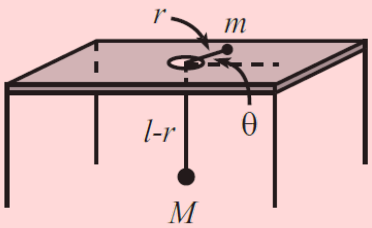
Amalie Emmy Noether
German mathematician/physicist

(23 March 1882 – 14 April 1935)

The professor who wasn't.

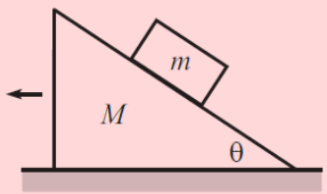
Lagrange's equations of motion

Problem 7. (Small oscillations) A mass m is free to slide on a frictionless table and is connected via a string that passes through a hole in the table to a mass M that hangs below. Assume that M moves in a vertical line only and assume that the string always remains taut.



(a) Find the equations of motion for the variables r and θ shown in the figure.
(b) Under what conditions does m undergo circular motion?
(c) What is the frequency of small oscillations (in the variable r) about this circular motion?

Problem 8. (Moving plane) A block of mass m is held motionless on a frictionless plane of mass M and angle of inclination θ. The plane rests on a frictionless horizontal surface.
The block is released.



(a) Derive the horizontal acceleration of the plane using Newton's laws.
(b) Derive the horizontal acceleration of the plane using the Lagrangian approach.